		Option C → ½(100)(10)^2 = 5000 J Option D → Atmospheric pressure is already 100 KPa
3	**C**	Equation of motion $v^2 = u^2 + 2as$ gives $\left(\frac{v_A}{3}\right)^2 = v_A^2 + 2(-9.81)(0.40)$ gives $v_A = 2.97 = 3.0\ \mathrm{m\ s^{-1}}$
4	**C**	Let velocity before collision be u, and after collision be V. Then conservation